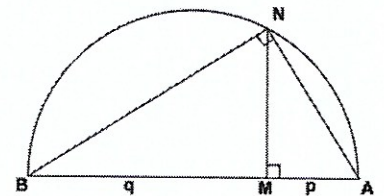


Inequalities WS 3

- Suppose that p, q and r are real and distinct.
 - Prove that $p^2 + q^2 > 2pq$
 - Hence prove that $p^2 + q^2 + r^2 > pq + qr + rp$.
 - Given that $p + q + r = 1$, prove that $pq + qr + rp < \frac{1}{3}$
- Suppose that a, b and c are real numbers.
 - Prove that $a^4 + b^4 + c^4 \geq a^2b^2 + a^2c^2 + b^2c^2$.
 - Hence show that $a^2b^2 + a^2c^2 + b^2c^2 \geq a^2bc + b^2ac + c^2ab$.
 - Deduce that if $a + b + c = d$, then $a^4 + b^4 + c^4 \geq abcd$.
- Suppose that a, b and c are positive.
 - Prove that $a^2 + b^2 \geq 2ab$.
 - Hence prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$
 - Given that $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$, prove that $a^3 + b^3 + c^3 \geq 3abc$.
 - If x, y and z are positive, show that $x + y + z > 3(xyz)^{\frac{1}{3}}$.
 - Suppose that $(1 + x)(1 + y)(1 + z) = 8$. Prove that $xyz \leq 1$.
- Suppose that a, b and c are positive.
 - Prove that $\frac{a}{b} + \frac{b}{a} \geq 2$
 - Hence show that $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$
 - Prove that $a^3 + b^3 \geq \left(\frac{a}{c} + \frac{b}{c} \right) abc$, and write down similar inequalities for $b^3 + c^3$ and $c^3 + a^3$.
 - Hence prove that $a^3 + b^3 + c^3 \geq 3abc$.
 - Deduce that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} > 3$.
- Suppose that x and y are positive. Prove that:
 - $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$
 - $\frac{1}{x^2} + \frac{1}{y^2} \geq \frac{8}{(x+y)^2}$
- In the diagram, AB is the diameter of a semicircle. $\angle ANB$ is 90° and M is a point on AB such that NM is perpendicular to AB . If $AM = p$ and $MB = q$
 - Show that $NM = \sqrt{pq}$
 - Deduce, using the diagram, that $\sqrt{pq} \leq \frac{p+q}{2}$
 - Use part b to prove that if $p, q, x, y > 0$, then $\frac{1}{4}(p + q + x + y) \geq (pqxy)^{\frac{1}{4}}$
 - Deduce that if $l, m, n, z > 0$, then $\frac{l}{m} + \frac{m}{n} + \frac{n}{z} + \frac{z}{l} \geq 4$



Answer

- Begin with $(p - q)^2 \geq 0$; b. Use part (a) three times and add; c. Begin with $(p + q + r)^2$ and use part (b).
- Use Question 1(b) with $p = a^2$ and so on; b. Use Question 1(b) with $p = ab$ and so on; c. Use part (a) and (b)
- See Question 1(a); b. See Question 1(b); c. Use part (b); d. Use part (c) and put $a^3 = x$ and so on; e. Expand and then used part (d). a more sophisticated approach is to use part (a) and in the first bracket put $a^2 = 1$ and $b^2 = x$, and so on.
- Use Question 3(a) and divide by ab ; b. Expand the LHS and use part (a); c. i. Begin with Question 3(a) and multiply by $(a + b)$; c.ii. Add part (i) and use part (a); c.iii. Begin with part (ii) and replace a^3 with $\frac{a}{b}$.
- Begin with Question 3(a) and put $a^2 = \frac{1}{x}$ and so on; b. Begin with Question 3(a) and put $a^2 = \frac{1}{x^2}$ and so on. Then combine with the square of part (a).